

Problem 1

Plot and use fsolve to find and mark roots of

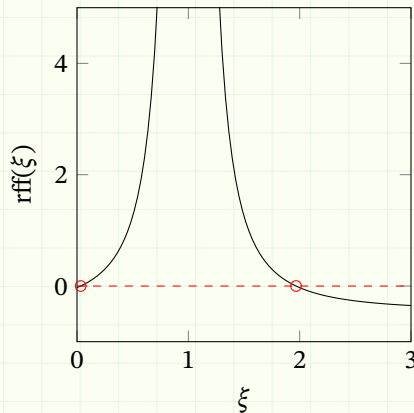
$$0.02946 = \frac{0.556\xi(2-\xi)}{[0.5(1-\xi)]^{1/2}[1.5(1-\xi)]^{3/2}}$$

Remember to rearrange the equation to rootfinding form as done in the local function below.

```

1 %% Script
2 xi1 = fsolve(@rff, 2); % solution near 2
3 xi2 = fsolve(@rff, 0.03); % solution near 0.03
4
5 xi = linspace(0, 3);
6 plot(xi, rff(xi), 'k-'); hold on;
7 plot([xi1, xi2], [0 0], 'ko'); hold off
8 yline(0, 'r--')
9 xlabel('\xi'); ylabel('rff(\xi)')
10 ylim([-1 5]); % helps to see function more clearly
11
12 %% Local function
13 function out = rff(xi)
14 numerator = 0.556*xi.*(2-xi);
15 denominator = sqrt(0.5*(1-xi)).*(1.5*(1-xi)).^(3/2);
16 out = numerator./denominator - 0.02946; % rootfinding form
17 end

```



The 0.02946 term can be moved to the RHS or the large fraction can be moved to the LHS to get the equation in RFF. The plot will look different but the roots will be the same.

The two roots are at $\xi = 1.9673$ and $\xi = 0.0327$. Were this an “actual” problem we’d note problems with the first answer because it will result in imaginary flow rates, but this problem didn’t specify anything other than “solve the equation for ξ ” so we accept both answers.

Problem 2

Solve simultaneously the equations

$$\xi_1 = 2.4 - 1.1\xi_2 - \sqrt{(2.4 - 1.1\xi_2)^2 - 1.6(2 - \xi_2)}$$

$$\xi_2 = -4 + \sqrt{16 + 4\xi_1(1 - \xi_1)}$$

Assume roots are between 0 and 1.

Roots are assumed to be between 0 and 1 so pick 0.5 as an initial guess for both. **Carefully** enter the equations in rootfinding form using the pattern from §1.8:

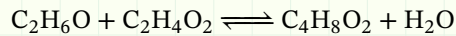
```
1 xi1Guess = 0.5;
2 xi2Guess = 0.5;
3 xiSol = fsolve(@rff, [xi1Guess;xi2Guess]);
4 fprintf('xi1 = %.4f, xi2 = %.4f\n', xiSol(1), xiSol(2));
5
6 function out = rff(xi)
7     xi1 = xi(1);
8     xi2 = xi(2);
9     eq1 = 2.4-1.1*xi2-sqrt((2.4-1.1*xi2).^2-1.6*(2-xi2))-xi1;
10    eq2 = -4+sqrt(16+4*xi1.*(1-xi1)) - xi2;
11    out = [eq1;eq2];
12 end
```

This results in $\xi_1 = 0.8041$ and $\xi_2 = 0.0780$. Check solution stability by trying a few different guess values: I tried 0.1 and 0.9 and got the same answer, so I'm confident it's stable.

I entered the equations incorrectly several times and got imaginary numbers when I ran fsolve!

Problem 3

Determine outlet mass flow rates of liquid-phase equilibrium reactor at 75 °C and 250 kPa for reaction



for feed of $1 \text{ kg} \cdot \text{min}^{-1}$ each of ethanol, acetic acid. Also determine conversion of ethanol.

It's good to get into the habit of making a stoichiometric table whenever you start a problem, so let's do that first. I'll also include the formation energies since those will be useful later:

Sp.	ν_i	n_{i0} (mol·min ⁻¹)	n_i (mol·min ⁻¹)	$\Delta H_{f,i}^\circ$ (kJ·mol ⁻¹)	$\Delta G_{f,i}^\circ$ (kJ·mol ⁻¹)
C ₂ H ₆ O	-1	21.7	$21.7 - \xi$	-277.7	-174.8
C ₂ H ₄ O ₂	-1	16.7	$16.7 - \xi$	-484.5	-389.9
C ₄ H ₈ O ₂	1	0.0	ξ	-463.3	-318.4
H ₂ O	1	0.0	ξ	-285.8	-237.1
	0	38.4	38.4	13.1	9.2

Determine ξ by using Eq. (2.32):

$$K_0 K_1 = \prod a_i^{\nu_i}$$

$$(0.0244)(2.1376) = \frac{x_C x_D}{x_A x_B}$$

$$0.0522 = \frac{\xi^2}{(21.7 - \xi)(16.7 - \xi)}$$

This is one equation with one unknown so fsolve can find ξ . Use a plot to get an estimate for ξ of about 3 mol·min⁻¹.

```
1 fplot(@rff, 'k', [0 6]) % helps to get guess
2 yline(0, 'r--')
3 xi = fsolve(@rff, 3) % find xi
4
5 function out = rff(xi)
6     out = xi.^2./((21.7-xi).*(16.7-xi))-0.0522;
7 end
```

From here, $\xi = 3.536 \text{ mol} \cdot \text{min}^{-1}$. Substituting back into n_i from the stoich table and then using molar masses to get mass flow rates provides the following solution:

Sp.	n_i (mol·min ⁻¹)	m_i (kg·min ⁻¹)
C ₂ H ₆ O	18.2	0.836
C ₂ H ₄ O ₂	13.2	0.790
C ₄ H ₈ O ₂	3.5	0.311
H ₂ O	3.5	0.064

Notice that pressure was never introduced: the reaction is liquid-phase, so pressure is (mostly) irrelevant.

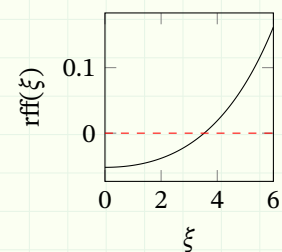
Sp.	MM
ethanol	46
acetic acid	60
ethyl acetate	88
water	18

$$K_0 = \exp \left[\frac{-9.2}{R(298)} \right] = 0.0244$$

$$K_1 = \exp \left[\frac{13.1}{R} \left(\frac{1}{298} - \frac{1}{T} \right) \right] = 2.1376$$

where $T = 75^\circ\text{C} = 348 \text{ K}$. Beware units! Make sure the arguments of the exponential are unitless (e.g., multiply energies by 1000 if you want to use $R = 8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$).

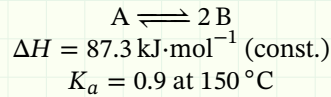
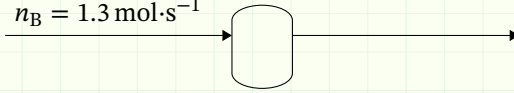
Let A = ethanol, B = acetic acid, C = ethyl acetate, D = water. Expressions for a_i from Table 2.3. Recall that $x_i = n_i/n_T$, and in the form Eq. (2.32) takes for this problem all n_T cancel out.



Problem 4

$$n_A = 0.8 \text{ mol}\cdot\text{s}^{-1}$$

$$n_B = 1.3 \text{ mol}\cdot\text{s}^{-1}$$



(A) K_a at 100°C .

(B) Outlet y_i for $T = 100^\circ\text{C}$, $P = 1.2 \text{ atm}$.

(C) X_A from 50°C to 300°C . Explain curve.

As usual we start with a stoichiometric table:

Sp.	ν_i	n_{i0} (mol·min ⁻¹)	n_i (mol·min ⁻¹)
A	-1	0.8	$0.8 - \xi$
B	2	1.3	$1.3 + 2\xi$
	1	2.1	$2.1 + \xi$

Part A

$$K_a = K_R \exp \left[\frac{\Delta H}{R} \left(\frac{1}{T_R} - \frac{1}{T} \right) \right]$$

$$= 0.9 \exp \left[\frac{87300}{8.314} \left(\frac{1}{423} - \frac{1}{373} \right) \right]$$

$$= \boxed{0.0323}$$

Remember to convert temperatures to kelvin!

Part B

Applying Eq. (2.32):

$$K_a = \prod a_i^{\nu_i}$$

$$0.0323 = \prod \left(y_i \frac{P}{P^\circ} \right)^{\nu_i}$$

$$0.0323 = \frac{y_B^2}{y_A} \frac{P}{P^\circ}$$

$$0.0323 = \frac{n_B^2}{n_A n_T} \frac{1.2}{1}$$

$$0.0323 = \frac{(1.3 + 2\xi)^2}{(0.8 - \xi)(2.1 + \xi)} \quad (1.2)$$

Reactor is gas phase so all $a_i = y_i P / P^\circ$ from Table 2.3.

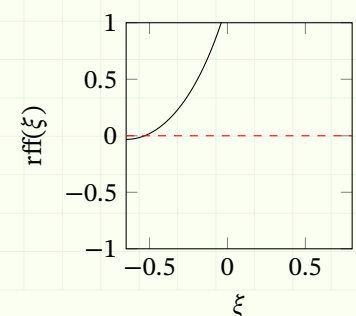
Convert pressure to consistent units. Since $P^\circ \equiv 101.325 \text{ kPa} = 1 \text{ atm}$ we could use atm or kPa. The former is a little cleaner to work with so it was chosen here.

which is one equation with one unknown. From the stoich table we know $\xi < 0.8$ (otherwise n_A will be negative) and $\xi > -1.3/2$ (otherwise n_B will be negative). A plot of Eq. (2.32) in RFF (see side panel) suggests a root near -0.5 , so we use that as the initial guess for fsolve:

```
1 xiSol = fsolve(@rff, 0.5)
2
3 function out = rff(xi)
4 out = (1.3+2*xi).^2./((0.8-xi).*(2.1+xi))*1.2-0.0323;
5 end
```

This provides $\xi = -0.53 \text{ mol}\cdot\text{s}^{-1}$. Substituting into the stoich table and then applying $y_i = n_i / n_T$ provides the outlet mole fractions of all species:

ξ can be positive or negative if reactants *and* products are present in the feed stream.



Sp.	ν_i	n_{i0} (mol·min ⁻¹)	n_i (mol·min ⁻¹)	y_i
A	-1	0.8	1.33	0.85
B	2	1.3	0.24	0.15
	1	2.1	1.57	

Part C

There are several ways to get the conversion of A at different temperatures but the key steps are:

1. Pick a temperature.
2. Calculate K_a .
3. Solve Eq. (2.32).
4. Repeat.

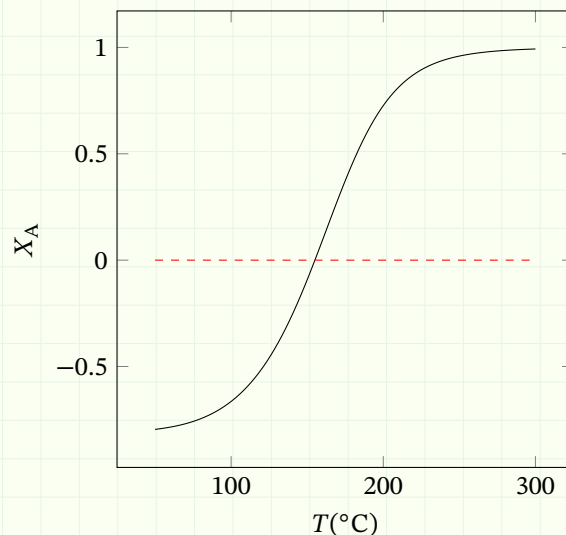
How we go about the third step can also vary: we could use our existing code to solve for ξ at various temperatures, or we could use the X_A formulation for each n_i from §2.8—specifically, Eq. (2.35)—and solve for X_A directly. Let's leave the existing code alone since we've already got it set up.

We can calculate K_a within the local function of our script and use the syntax from §1.7 to pass each new temperature as an additional input argument (see Example 1.7). Based on the plot in Part B it looks like the rootfinding function is fairly well behaved so we'll pick a guess of 0 and hope that it works for all various temperatures (we'll know it goes wrong if the plot looks odd, or if fsolve prints something worrisome to the Command Window).

```

1 T = linspace(50,300)+273;
2 xiSol = zeros(size(T));
3 for i = 1:length(T)
4     xiSol(i) = fsolve(@(xi)rff(xi,T(i)), 0);
5 end
6 Xa = 1-(0.8-xiSol)/0.8; % eq. (2.33)
7 plot(T-273, Xa)
8
9 function out = rff(xi, T)
10     Ka = 0.9*exp(87300/8.314*(1/423-1/T));
11     out = (1.3+2*xi).^2./((0.8-xi).*(2.1+xi))*1.2-Ka;
12 end

```



The reaction is *endothermic* ($\Delta H > 0$) so we expect high temperatures to favor product formation. As shown in the plot at high temperatures the net reaction proceeds *forwards* as $A \longrightarrow 2 B$ (conversion of A is positive). However at low temperatures the net reaction proceeds *backwards* as $2 B \longrightarrow A$ (conversion of A is negative). The temperature at which the net reaction does not occur is about 155°C and corresponds to the temperature at which the equilibrium composition is equal to the feed composition.

Problem 5

Problem 4 reaction also allowing $2\text{B} \rightleftharpoons \text{C}$ with $K_a = 1.1$ at 90°C , $\Delta H = -62.9\text{ kJ}\cdot\text{mol}^{-1}$. Get n_i for reactor at 100°C , 1.2 atm.

A stoich table to help us get started:

Sp.	ν_{ij}		n_{i0} (mol·min ⁻¹)	n_i (mol·min ⁻¹)
A	-1	0	0.8	$0.8 - \xi_1$
B	2	-2	1.3	$1.3 + 2\xi_1 - 2\xi_2$
C	0	1	0.0	ξ_2
	1	-1	2.1	$2.1 + \xi_1 - \xi_2$

We must now solve simultaneously two instances of Eq. (2.32), once for the first reaction and once for the second reaction:

$$0.0323 = 1.2 \frac{n_B^2}{n_A n_T} \quad (\text{Reaction 1})$$

$$0.6291 = \frac{1}{1.2} \frac{n_C n_T}{n_B^2} \quad (\text{Reaction 2})$$

$$K_{a,2} = 1.1 \exp \left[\frac{-62900}{8.314} \left(\frac{1}{363} - \frac{1}{373} \right) \right]$$

from which $K_{a,2} = 0.6291$.

In these equations all n_i terms are defined in the stoichiometric table, which means we have two equations (two instances of Eq. (2.32)) and two unknowns (ξ_1 and ξ_2). Using initial guesses for both of 0.5, fsolve can be used as follows:

```

1 xiSol = fsolve(@rff, [0.5, 0.5])
2
3 function out = rff(xi)
4     xi1 = xi(1);
5     xi2 = xi(2);
6
7     nA = 0.8-xi1;
8     nB = 2*xi1-2*xi2+1.3;
9     nC = xi2;
10    nT = xi1-xi2+2.1;
11
12    eq1 = 1.2*nB.^2./(nA.*nT) - 0.0323;
13    eq2 = (1/1.2)*nC.*nT./(nB.^2) - 0.6291;
14
15    out = [eq1; eq2];
16 end

```

This code provides $\xi_1 = -0.5061$ (as $\text{xi1} = \text{xiSol}(1)$) and $\xi_2 = 0.0265$ (as $\text{xi2} = \text{xiSol}(2)$). The output molar flow rates of all species are calculated by substituting these values back into the stoichiometric table:

Sp.	ν_{ij}		n_{i0} (mol·min ⁻¹)	n_i (mol·min ⁻¹)
A	-1	0	0.8	1.306
B	2	-2	1.3	0.235
C	0	1	0.0	0.027
	1	-1	2.1	1.567